

```
# Cell 1 – Install libraries and download models/corpora (run once)
!pip install -U spacy textblob
!python -m spacy download en_core_web_sm
!python -m textblob.download_corpora
```

Requirement already satisfied: catalogue<21.0, >=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (20.12.0)
Requirement already satisfied: weasel<0.5.0, >=0.1.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.1)
Requirement already satisfied: typer<1.0.0, >=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.17.4)
Requirement already satisfied: tqdm<5.0.0, >=4.38.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (4.67.1)
Requirement already satisfied: numpy<=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
Requirement already satisfied: requests<3.0.0, >=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
Requirement already satisfied: pydantic!=1.8, !=1.8.1, <3.0.0, >=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.10.6)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.2.0)
Requirement already satisfied: packaging<=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (25.0)
Requirement already satisfied: langcodes<4.0.0, >=3.2.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.5.0)
Requirement already satisfied: nltk<=3.9 in /usr/local/lib/python3.12/dist-packages (from textblob) (3.9.1)
Requirement already satisfied: language-data<=1.2 in /usr/local/lib/python3.12/dist-packages (from langcodes<4.0.0, >=3.2.0->spacy) (1.2.0)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk<=3.9->textblob) (8.2.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk<=3.9->textblob) (1.5.2)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk<=3.9->textblob) (2024.11.11)
Requirement already satisfied: annotated-types<=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8, !=1.8.1, <3.0.0, >=1.7.4->spacy) (0.7.0)
Requirement already satisfied: pydantic-core<=2.33.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8, !=1.8.1, <3.0.0, >=1.7.4->spacy) (2.33.2)
Requirement already satisfied: typing-extensions<=4.12.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8, !=1.8.1, <3.0.0, >=1.7.4->spacy) (4.12.2)
Requirement already satisfied: typing-inspection<=0.4.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8, !=1.8.1, <3.0.0, >=1.7.4->spacy) (0.4.0)
Requirement already satisfied: charset-normalizer<4, >=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0, >=2.13.0->spacy) (3.4.0)
Requirement already satisfied: idna<4, >=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0, >=2.13.0->spacy) (3.10.1)
Requirement already satisfied: urllib3<3, >=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0, >=2.13.0->spacy) (2.3.0)
Requirement already satisfied: certifi<=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0, >=2.13.0->spacy) (2019.9.11)
Requirement already satisfied: blis<1.4.0, >=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0, >=8.3.4->spacy) (1.3.0)
Requirement already satisfied: confection<1.0.0, >=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0, >=8.3.4->spacy) (0.0.4)
Requirement already satisfied: shellingham<=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0.0, >=0.3.0->spacy) (1.5.4)
Requirement already satisfied: rich<=10.11.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0.0, >=0.3.0->spacy) (13.9.0)
Requirement already satisfied: cloudpathlib<1.0.0, >=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0, >=0.1.0->spacy) (0.18.0)
Requirement already satisfied: smart-open<8.0.0, >=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0, >=0.1.0->spacy) (7.0.4)
Requirement already satisfied: MarkupSafe<=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->spacy) (3.0.2)
Requirement already satisfied: marisa-trie<=1.1.0 in /usr/local/lib/python3.12/dist-packages (from language-data<=1.2->langcodes<4.0.0, >=3.2.0->spacy) (1.1.0)
Requirement already satisfied: markdown-it-py<=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich<=10.11.0->typer<1.0.0, >=0.3.0->spacy) (3.0.0)
Requirement already satisfied: pygments<3.0.0, >=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich<=10.11.0->typer<1.0.0, >=0.3.0->spacy) (2.19.1)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open<8.0.0, >=5.2.1->weasel<0.5.0, >=0.1.0->spacy) (1.16.0)
Requirement already satisfied: mdurl<=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py<=2.2.0->rich<=10.11.0->typer<1.0.0, >=0.3.0->spacy) (0.1.2)
Collecting en-core-web-sm<=3.8.0
 Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl 12.8/12.8 MB 89.0 MB/s eta 0:00:00

✓ Download and installation successful

You can now load the package via `spacy.load('en_core_web_sm')`

⚠ Restart to reload dependencies

If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the 'Restart kernel' or 'Restart runtime' option.

```
[nltk_data] Downloading package brown to /root/nltk_data...
[nltk_data] Unzipping corpora/brown.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Downloading package averaged_perceptron_tagger_eng to /root/nltk_data...
[nltk_data] Unzipping taggers/averaged_perceptron_tagger_eng.zip.
[nltk_data] Downloading package conll12000 to /root/nltk_data...
[nltk_data] Unzipping corpora/conll12000.zip.
[nltk_data] Downloading package movie_reviews to /root/nltk_data...
[nltk_data] Unzipping corpora/movie_reviews.zip.
Finished.
```

Cell 2 – Imports and load spaCy model

```
import spacy
from textblob import TextBlob

nlp = spacy.load("en_core_web_sm")
```

Optional Cell – Download NLTK punkt if you see tokenizer errors

```
import nltk
nltk.download('punkt')
nltk.download('wordnet')
nltk.download('averaged_perceptron_tagger')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to /root/nltk_data...
[nltk_data] Unzipping taggers/averaged_perceptron_tagger.zip.
True
```

```
# Cell 3 – Text, Named Entity Recognition (NER) and Sentiment with TextBlob
text = "Amazon is acquiring a startup in San Francisco for $1 billion. I am excited about this news!"

# --- NER ---
doc = nlp(text)
print("\n=== Named Entities ===")
for ent in doc.ents:
    print(f"Text: {ent.text}, Label: {ent.label_}")

# --- Sentiment ---
blob = TextBlob(text)
print("\n=== Sentiment ===")
print("Polarity:", blob.sentiment.polarity)
print("Subjectivity:", blob.sentiment.subjectivity)
```

```
=== Named Entities ===
Text: Amazon, Label: ORG
Text: San Francisco, Label: GPE
Text: $1 billion, Label: MONEY

=== Sentiment ===
Polarity: 0.46875
Subjectivity: 0.75
```

Start coding or [generate](#) with AI.